Discrete Random Variables & Binomial Distributions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Probability distributions and expected value**

A game costs 2 dollars to play. You win 10 dollars with probability 0.1, and win nothing otherwise. Let X be your net profit. Find the probability distribution of X and its expected value.

$X = 8$ (win 10, minus the 2 cost) with probability 0.1; $X = -2$ (lose the 2 cost) with probability 0.9. Expected value: $E(X) = 8(0.1) + (-2)(0.9) = 0.8 - 1.8 = -1$. On average, a player loses 1 dollar per game.

2 **[Foundational] • Target: Probability distributions and expected value**

Find the standard deviation of the profit X from the previous problem, to two decimal places.

Variance $= \sum (x-\mu)^2 P(x) = (8-(-1))^2(0.1) + (-2-(-1))^2(0.9) = (81)(0.1) + (1)(0.9) = 8.1 + 0.9 = 9$. Standard deviation $= \sqrt{9} = 3$.

3 **[Foundational] • Target: Binomial setting and conditions**

State the four conditions required for a random variable to follow a binomial distribution (sometimes remembered as BINS: Binary, Independent, Number fixed, Same probability).

(1) Each trial has only two possible outcomes (success/failure). (2) Trials are independent. (3) There is a fixed number of trials, n. (4) The probability of success, p, is the same for every trial.

4 **[Foundational] • Target: Binomial setting and conditions**

A survey randomly selects 20 people (without replacement) from a small town of 50 people and records how many support a proposal. Explain why this technically violates the independence condition, and why it is often still treated as approximately binomial.

Sampling without replacement from a small population means each selection slightly changes the probabilities for the next draw, violating strict independence. However, when the sample size is small relative to the population (a common guideline is the sample being less than 10% of the population), the effect is negligible, and the situation is treated as approximately binomial — though here 20 out of 50 is 40%, so this would NOT satisfy the 10% condition and should be treated with caution.

5 **[Foundational] • Target: Binomial probability formula**

A basketball player makes 70% of free throws. Find the probability she makes exactly 4 out of 6 attempts, using $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$.

$P(X = 4) = \binom{6}{4}(0.7)^4(0.3)^2 = 15(0.2401)(0.09) \approx 0.324$.

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6 [Foundational] • Target: Binomial probability formula

Using the same free-throw shooter, find the mean and standard deviation of the number of makes out of 6 attempts, using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$.

$\mu = np = 6(0.7) = 4.2$ makes. $\sigma = \sqrt{np(1-p)} = \sqrt{6(0.7)(0.3)} = \sqrt{1.26} \approx 1.12$ makes.

7 [Foundational] • Target: Visualizing a binomial distribution

The graph shows a smoothed approximation of the binomial distribution for $n = 20$ trials with $p = 0.5$, plotted as a continuous curve for visual comparison to a normal shape.

a Using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$, verify that this binomial distribution has mean 10 and standard deviation approximately 2.24, matching the curve shown. $\mu = np = 20(0.5) = 10$ ✓. $\sigma = \sqrt{np(1-p)} = \sqrt{20(0.5)(0.5)} = \sqrt{5} \approx 2.24$ ✓.

b Explain why a binomial distribution with $n = 20$ and $p = 0.5$ looks approximately bell-shaped, referencing the conditions for a normal approximation. A binomial distribution can be approximated by a normal distribution when both $np \geq 10$ and $n(1-p) \geq 10$. Here $np = 10$ and $n(1-p) = 10$, both meeting the threshold, so the shape is approximately bell-shaped/symmetric rather than skewed.

(a) $\mu = np = 20(0.5) = 10$ ✓. $\sigma = \sqrt{np(1-p)} = \sqrt{20(0.5)(0.5)} = \sqrt{5} \approx 2.24$ ✓.; (b) A binomial distribution can be approximated by a normal distribution when both $np \geq 10$ and $n(1-p) \geq 10$. Here $np = 10$ and $n(1-p) = 10$, both meeting the threshold, so the shape is approximately bell-shaped/symmetric rather than skewed.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric distribution (brief introduction)

A basketball player makes free throws with probability 0.7 each attempt, independently. Find the probability that her first miss occurs on the 3rd attempt (i.e. she makes the first 2, then misses the 3rd).

$P(\text{first miss on attempt } 3) = (0.7)(0.7)(0.3) = 0.147$, using the geometric distribution's structure (successes then the first failure).

9 [Intermediate] • Target: Applying the binomial distribution in context

A multiple-choice quiz has 10 questions, each with 4 choices, and a student guesses randomly on every question. Find the probability the student gets at least 8 questions correct, given that $P(X = 8) \approx 0.0004$, $P(X = 9) \approx 0.00003$, and $P(X = 10) \approx 0.000001$.

$P(X \geq 8) = P(X = 8) + P(X = 9) + P(X = 10) \approx 0.0004 + 0.00003 + 0.000001 \approx 0.00043$, or about 0.043% — an extremely small chance, illustrating how unlikely it is to do well on a multiple-choice test by pure guessing.

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10 **[Intermediate]** • **Target: Binomial probability of at least one success**

A manufacturing process produces defective items independently with probability 0.05. In a batch of 20 items, find the probability that at least one item is defective, using the complement of "zero defective."

$P(\text{zero defective}) = (0.95)^{20} \approx 0.358$. $P(\text{at least one defective}) = 1 - 0.358 = 0.642$.

11 **[Intermediate]** • **Target: Binomial probability of at least one success**

Using the mean and standard deviation formulas $\mu = np$ and $\sigma = \sqrt{np(1-p)}$, find the mean and standard deviation of the number of defective items in a batch of 20, with $p = 0.05$.

$\mu = np = 20(0.05) = 1$. $\sigma = \sqrt{np(1-p)} = \sqrt{20(0.05)(0.95)} = \sqrt{0.95} \approx 0.975$.

12 **[Intermediate]** • **Target: Comparing binomial scenarios**

Explain why a binomial model would NOT be appropriate for counting the number of aces dealt when drawing 5 cards without replacement from a standard 52-card deck, and name the distribution that would be more appropriate instead.

Drawing without replacement means each draw changes the composition of the remaining deck, so the probability of drawing an ace changes from draw to draw — violating the binomial requirement of a constant probability of success on every trial. This scenario is better modeled by the hypergeometric distribution, which accounts for sampling without replacement from a finite population.

13 **[Intermediate]** • **Target: Geometric distribution (brief introduction)**

A discrete random variable X has probability distribution: $P(X=0) = 0.1$, $P(X=1) = 0.3$, $P(X=2) = 0.4$, $P(X=3) = 0.2$. Calculate: (a) $E(X)$, (b) $E(X^2)$, and (c) $\text{Var}(X)$ and σ_X.

(a) Expected Value: $E(X) = 0(0.1) + 1(0.3) + 2(0.4) + 3(0.2) = 0 + 0.3 + 0.8 + 0.6 = \mathbf{1.7}$. (b) Expected Value of X^2: $E(X^2) = 0^2(0.1) + 1^2(0.3) + 2^2(0.4) + 3^2(0.2) = 0 + 0.3 + 1.6 + 1.8 = \mathbf{3.7}$. (c) Variance & SD: $\text{Var}(X) = E(X^2) - [E(X)]^2 = 3.7 - (1.7)^2 = 3.7 - 2.89 = \mathbf{0.81}$. $\sigma_X = \sqrt{0.81} = \mathbf{0.9}$.

14 **[Intermediate]** • **Target: Applying the binomial distribution in context**

Independent random variables X and Y have $\mu_X = 50$, $\sigma_X = 6$ and $\mu_Y = 30$, $\sigma_Y = 8$. Define $W = 3X - 2Y + 10$. Find the mean μ_W and standard deviation σ_W.

Mean: $\mu_W = 3\mu_X - 2\mu_Y + 10 = 3(50) - 2(30) + 10 = 150 - 60 + 10 = \mathbf{100}$. Variance (Variances Always Add!): Adding constants adds zero variance. Coefficients are squared: $\sigma_W^2 = 3^2\sigma_X^2 + (-2)^2\sigma_Y^2 = 9(6^2) + 4(8^2) = 9(36) + 4(64) = 324 + 256 = 580$. Standard Deviation: $\sigma_W = \sqrt{580} \approx \mathbf{24.08}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Binomial probability of at least one success • Cross-Lesson: 1.01 Measures of Center & Spread

Level 1 · FoundationsThe temperatures (in °C) during a science experiment have median $M = 22.0^\circ\text{C}$ and $\text{IQR} = 4.5^\circ\text{C}$. Convert the temperatures to Fahrenheit using the formula $F = 1.8C + 32$. What are the median and IQR of the temperatures in °F?

Median in °F: Scaled by 1.8 and shifted by 32: $M_F = 1.8(22.0) + 32 = 39.6 + 32 = \mathbf{71.6}^\circ\text{F}$. IQR in °F: Spread is affected ONLY by the multiplicative factor 1.8, not the addition of 32: $\text{IQR}_F = 1.8(\text{IQR}_C) = 1.8(4.5) = \mathbf{8.1}^\circ\text{F}$.

16 [Challenging] • Target: Comparing binomial scenarios • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.

(a) In a 2-week period, the parameter scales to $\lambda = 1.6 \times 2 = 3.2$: $P(X \geq 3) = 1 - P(X \leq 2) = 1 - e^{-3.2}\left(1 + 3.2 + \frac{3.2^2}{2}\right) = 1 - 9.32e^{-3.2} \approx 1 - 0.3799 = 0.620$ (3 s. f.) (b) For 1 week, $\lambda = 1.6$. Use conditional probability: $P(X = 2 \mid X \geq 1) = \frac{P(X=2)}{P(X \geq 1)} = \frac{\frac{e^{-1.6}(1.6)^2}{2}}{1 - e^{-1.6}} = \frac{1.28e^{-1.6}}{1 - e^{-1.6}} \approx \frac{0.2584}{0.7981} \approx 0.324$ (3 s. f.)

17 [Challenging] • Target: Probability distributions and expected value • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A discrete random variable X follows a binomial distribution $X \sim B(15, 0.35)$. (a) Find $P(X = 5)$. (b) Find $P(X \geq 4)$. (c) Calculate the mean $E(X)$ and standard deviation σ of X.

(a) $P(X = 5) = \binom{15}{5}(0.35)^5(0.65)^{10} = 3003 \times (0.005252) \times (0.013463) \approx 0.2123$. (b) $P(X \geq 4) = 1 - P(X \leq 3)$. Using GDC `binomcdf(15, 0.35, 3)`: $P(X \leq 3) \approx 0.1727$. $P(X \geq 4) = 1 - 0.1727 = 0.8273 \quad (82.7\%)$. (c) Mean: $E(X) = np = 15 \times 0.35 = 5.25$. Variance: $\text{Var}(X) = np(1-p) = 15 \times 0.35 \times 0.65 = 3.4125$. Standard deviation: $\sigma = \sqrt{3.4125} \approx 1.847 \approx 1.85$.

18 [Challenging] • Target: Binomial setting and conditions • Cross-Lesson: 2.01 Sampling Methods & Bias

Practice 4.1: A multiple-choice examination consists of 20 independent questions, each having 4 options of which only one is correct. A student guesses randomly on all questions. Let X be the number of correct answers. (a) State the distribution of X including its parameters. (b) Find the probability that the student passes by scoring at least 10 correct answers. (c) Calculate the mean and standard deviation of X.

(a) $X \sim B(n, p)$ with $n = 20$ and $p = 0.25$. (b) $P(X \geq 10) = 1 - P(X \leq 9)$. Using GDC `binomcdf(20, 0.25, 9)`: $P(X \leq 9) \approx 0.9861$. $P(X \geq 10) = 1 - 0.9861 = 0.0139 \quad (1.39\%)$. (c) Mean: $\mu = np = 20 \times 0.25 = 5$. Standard deviation: $\sigma = \sqrt{np(1-p)} = \sqrt{20 \times 0.25 \times 0.75} = \sqrt{3.75} \approx 1.936 \approx 1.94$.

19 [Challenging] • Target: Binomial probability formula • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Cars arrive at a toll booth at an average rate of 3 per minute. Find the probability that (a) exactly 5 arrive in one minute, (b) at least 2 arrive in 30 seconds.

(a) $P(X = 5) = \frac{e^{-3} \cdot 243}{120} \approx 0.1008$. (b) $\lambda = 1.5$: $P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-1.5} - 1.5e^{-1.5} = 1 - 2.5e^{-1.5} \approx 0.4422$.

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20 **[Challenging]** • **Target: Visualizing a binomial distribution** • **Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A diagnostic medical screening test is 95% sensitive (positive if disease present) and 90% specific (negative if disease absent). In a population, 2% of people have the disease. Find the probability that a person who tests positive actually has the disease.

Let D = disease present, T^+ = positive test. $P(D) = 0.02$, $P(D^c) = 0.98$. Sensitivity: $P(T^+ \mid D) = 0.95$. Specificity: $P(T^- \mid D^c) = 0.90 \Rightarrow P(T^+ \mid D^c) = 0.10$. By the Law of Total Probability:

$P(T^+) = P(D)P(T^+ \mid D) + P(D^c)P(T^+ \mid D^c) = (0.02)(0.95) + (0.98)(0.10) = 0.019 + 0.098 = 0.117.$ By Bayes' Theorem:

$P(D \mid T^+) = \frac{P(D \cap T^+)}{P(T^+)} = \frac{0.019}{0.117} \approx \mathbf{0.1624}$ (16.24%). Despite 95% sensitivity, only ~16.2% of positive testers.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: Probability distributions and expected value**

A binomial random variable has $n = 14$ trials with success probability $p = 0.7$. Find the mean $\mu = np$.

a. 4.2

b. 10.8

c. 8.8

d. 9.8 ✓

$\mu = np = 14(0.7) = 9.8$.

22 **[Foundational]** • **Target: Binomial setting and conditions**

A binomial random variable has $n = 15$ trials with success probability $p = 0.9$. Find the standard deviation $\sigma = \sqrt{np(1-p)}$, to two decimal places.

a. 1.35

b. 1.16 ✓

c. 0.86

d. 1.46

$\sigma = \sqrt{np(1-p)} = \sqrt{15(0.9)(0.1)} = \sqrt{1.35} \approx 1.16$.

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23 [Foundational] • Target: Binomial probability formula

A manufacturing process produces defective items independently with probability 0.3. In a batch of 15 items, find the probability that at least one item is defective, using the complement of "zero defective," to three decimal places.

a. 1.045

b. 0.995 ✓

c. 0.005

d. 0.945

P(zero defective) $= (0.7)^{15} \approx 0.005$. P(at least one) $= 1 - 0.005 = 0.995$.

24 [Foundational] • Target: Visualizing a binomial distribution

A process succeeds independently with probability $p = 0.5$ on each of $n = 10$ trials. Find the probability of exactly 9 successes, using $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, to three decimal places.

a. 0.001

b. 0.020

c. 0.060

d. 0.010 ✓

$P(X = 9) = \binom{10}{9}(0.5)^9(0.5)^1 = 10(0.5)^9(0.5)^1 \approx 0.010$.

25 [Foundational] • Target: Geometric distribution (brief introduction)

A game costs 5 dollars to play. You win 15 dollars with probability 0.9, and win nothing otherwise. Letting X be your net profit, find $E(X)$.

a. $-\frac{7}{2}$

b. $\frac{15}{2}$

c. $\frac{17}{2}$ ✓

d. $\frac{19}{2}$

$X = 10$ (win, minus the 5 cost) with probability 0.9; $X = -5$ (lose the cost) with probability 0.1.
$E(X) = 10(0.9) + (-5)(0.1) = \frac{17}{2}$.

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26 [Foundational] • Target: Applying the binomial distribution in context

Binomial distribution requires:

a. Continuous outcomes

b. **Fixed n, two outcomes, constant p, independence ✓**

c. Normal data

d. Dependent trials

BINS: Binary, Independent, fixed N, same probability p.

27 [Foundational] • Target: Binomial probability of at least one success

A binomial random variable has $n = 6$ trials and success probability $p = 0.25$. Find $P(X = 6)$.

a. 0.0044

b. **0.0002 ✓**

c. 0.1318

d. 0.033

$P(X = 6) = \binom{6}{6}(0.25)^6(1 - 0.25)^0 = 0.0002$.

28 [Foundational] • Target: Comparing binomial scenarios

A geometric random variable counts trials until the first success, with success probability $p = 0.29$ each trial. Find $P(X = 6)$, the probability the first success occurs on trial 6.

a. 0.0737

b. 0.1462

c. **0.0523 ✓**

d. 0.0264

$P(X = 6) = (1 - 0.29)^5(0.29) = 0.0523$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Probability distributions and expected value

A binomial random variable has $n = 30$ and $p = 0.24$. Find its standard deviation $\sigma = \sqrt{np(1-p)}$.

a. **2.339 ✓**

b. 0.339

c. 1.839

d. 1.339

$\sigma = \sqrt{30 \times 0.24 \times 0.76} = 2.339$.

30 [Intermediate] • Target: Binomial setting and conditions

Normal distribution is symmetric about:

a. Standard deviation

b. Median only

c. **The mean ✓**

d. Zero

The normal curve is symmetric about its mean μ.

31 [Intermediate] • Target: Binomial probability formula

The Central Limit Theorem states that sample means are approximately Normal when:

a. Population is Normal

b. **n is large (n≥30) ✓**

c. σ is known

d. p<0.5

CLT: for large n, $\bar{x}$ is approximately Normal regardless of population shape.

32 [Intermediate] • Target: Visualizing a binomial distribution

A Normal random variable has $\mu = 189$ and $\sigma = 8$. Find $P(173 < X < 213)$.

a. 0.8759

b. **0.9759 ✓**

c. 0.7759

d. 0.6759

$z_1 = \frac{173-189}{8} = -2$, $z_2 = \frac{213-189}{8} = 3$. $P(173 < X < 213) = 0.9759$.

33 [Intermediate] • Target: Geometric distribution (brief introduction)

Events A and B satisfy $P(A) = 0.55$, $P(B) = 0.12$, and $P(A \cap B) = 0.0$. Find $P(A \cup B)$.

a. 0.57

b. **0.67 ✓**

c. 0.82

d. 0.62

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.55 + 0.12 - 0.0 = 0.67$.

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34 [Intermediate] • Target: Applying the binomial distribution in context

P(A and B) if A,B independent:

a. $P(A) + P(B)$

b. $P(A) \cdot P(B)$ ✓

c. $P(A|B)$

d. $P(A) + P(B) - P(A \cup B)$

Independence: multiply probabilities.

35 [Intermediate] • Target: Binomial probability of at least one success

In a standard Normal, what % of values fall within 1σ?

a. **~68%** ✓

b. ~95%

c. ~99.7%

d. ~50%

68-95-99.7 rule: 68% within ±1σ.

36 [Intermediate] • Target: Comparing binomial scenarios

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

a. **2.4** ✓

b. 3.6

c. 6.4

d. 7.2

For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$ and the variance is given by $Var(X) = np(1 - p)$. We are given that $n = 10$ and $E(X) = 4$. Using the formula for the mean, we have $10p = 4$. Solving for p, we get $p = \frac{4}{10} = 0.4$. Now we can find the variance using the formula $Var(X) = np(1 - p)$. Substituting the values of n and p, we get $Var(X) = 10 \times 0.4 \times (1 - 0.4)$. $Var(X) = 10 \times 0.4 \times 0.6$. $Var(X) = 4 \times 0.6$. $Var(X) = 2.4$.

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37 **[Intermediate]** • **Target: Probability distributions and expected value**

A researcher is designing a survey to understand student preferences for extracurricular activities at a large university. The university has 4 distinct colleges: Arts & Sciences, Engineering, Business, and Health Sciences. The researcher decides to use stratified random sampling. If each college represents one stratum, and the researcher wants to survey 200 students in total, with the number of students surveyed from each college being proportional to the college's enrollment, which of the following represents a valid way to determine the sample size for each college?

a. A simple random sample of 200 students from the entire university population.

b. A systematic sample of 200 students from an alphabetized list of all students.

c. A convenience sample of 200 students from the campus cafeteria.

d. **Proportionally allocating the 200 students to each of the 4 colleges based on their enrollment figures.** ✓

Stratified random sampling involves dividing the population into subgroups (strata) and then taking a random sample from each stratum. In this scenario, the 4 colleges are the strata. Proportional allocation means that the number of students sampled from each college is in proportion to that college's size relative to the total university enrollment. Therefore, allocating the 200 students proportionally to the enrollment of each of the 4 colleges is the correct approach for stratified random sampling.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Binomial setting and conditions • Cross-Lesson: 2.01 Sampling Methods & Bias

A histogram is used to display the distribution of the number of hours college students reported spending on social media per day. The bins for the histogram are as follows: 0-2 hours, 2-4 hours, 4-6 hours, 6-8 hours, and 8-10 hours. The heights of the bars are 5, 12, 15, 8, and 3, respectively. What is the total number of students represented by this histogram?

a. 15

b. 43

c. **50 ✓**

d. 200

The heights of the bars in a histogram represent the frequencies (counts) of data points within each bin. To find the total number of students represented, we sum the heights of all the bars. Total students = Height of bar 1 + Height of bar 2 + Height of bar 3 + Height of bar 4 + Height of bar 5 Total students = 5 + 12 + 15 + 8 + 3 = 43. Therefore, 43 students are represented by this histogram.

39 [Challenging] • Target: Binomial probability formula • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A distribution is displayed as a symmetric, bell-shaped histogram with a single peak in the middle. How should this shape be described?

a. Bimodal

b. Skewed left (negatively skewed)

c. Skewed right (positively skewed)

d. **Approximately symmetric (unimodal) ✓**

This description matches the shape: Approximately symmetric (unimodal). In this case, mean ≈ median.

40 [Challenging] • Target: Visualizing a binomial distribution • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A distribution is displayed as a histogram with two separate peaks. How should this shape be described?

a. Skewed right (positively skewed)

b. Approximately symmetric (unimodal)

c. Skewed left (negatively skewed)

d. **Bimodal ✓**

This description matches the shape: Bimodal. In this case, no simple mean/median rule applies.

41 [Challenging] • Target: Geometric distribution (brief introduction) • Cross-Lesson: 3.03 The Normal Distribution

A distribution is right-skewed when:

a. Mean < Median

b. **Mean > Median ✓**

c. Mean = Median

d. Mode > Mean

Right skew pulls mean above median.

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42 [Challenging] • Target: Applying the binomial distribution in context • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2 - X)$ m, where X is a continuous random variable with probability density function $f(x) = \begin{cases} \frac{3}{8}x^2, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$ Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

a. $\frac{127}{351}$ ✓

b. $\frac{74}{351}$

c. $\frac{37}{351}$

d. $\frac{127}{640}$

Let A denote the area of the rectangle and $L_{\min}$ denote the length of the shorter side. 1. **Determine the condition for $A \ge \frac{3}{4}$:** The area is given by $A = X(2 - X) = 2X - X^2$.
$2X - X^2 \ge \frac{3}{4} \Leftrightarrow X^2 - 2X + \frac{3}{4} \le 0 \Leftrightarrow (X - \frac{1}{2})(X - \frac{3}{2}) \le 0$ Thus, $A \ge \frac{3}{4}$ corresponds to the interval $E_1 = [\frac{1}{2}, \frac{3}{2}]$. Using the cumulative distribution function.

43 [Challenging] • Target: Binomial probability of at least one success • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

The bar graph below shows the distribution of the number of siblings reported by the 25 students in a homeroom class. If 1 student is selected at random from the class, what is the probability that the selected student has at least 2 siblings?

a. $\frac{6}{25}$

b. $\frac{9}{25}$ ✓

c. $\frac{12}{25}$

d. $\frac{16}{25}$

To find the probability that a randomly chosen student has at least 2 siblings: 1. Find the total number of outcomes (students): $6 + 10 + 6 + 3 = 25$ 2. Find the number of favorable outcomes. The phrase "at least 2 siblings" means having 2 or more siblings (either 2 or 3 siblings): - Number of students with 2 siblings: 6 - Number of students with 3 siblings: 3 - Favorable outcomes: $6 + 3 = 9$ 3. Calculate the probability: $P(\text{at least 2 siblings}) = \frac{9}{25}$ Common Misconceptions.

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44 [Challenging] • Target: Comparing binomial scenarios • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

Which of the following conditions is necessary to establish a cause-and-effect relationship between two variables?

a. A large sample size greater than 10,000

b. Random selection of participants from the population

c. **Random assignment of participants to experimental treatment groups ✓**

d. A correlation coefficient of exactly 1.0

Random assignment of treatments is the critical requirement that controls for confounding variables and enables causal inference.

45 [Challenging] • Target: Probability distributions and expected value • Cross-Lesson: 1.01 Measures of Center & Spread

In $\hat{y} = a + bx$, the slope b represents:

a. The predicted value at x=0

b. **Change in y per unit x ✓**

c. Correlation coefficient

d. Standard error

Slope = rise over run; change in ■ per unit change in x.

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